

Sanjeevraj Bommannan M

Embedded Systems Engineer | Electronics & Communication Engineering Undergraduate



Electronics and Communication Engineering undergraduate with a strong interest in Embedded Systems, IoT, and Microcontroller based applications. Experienced in developing embedded projects using ESP32, Arduino, 8051, sensors, and embedded C/C++. Skilled in hardware interfacing, sensor integration, serial communication, and real-time monitoring systems. Seeking an Embedded Systems Engineer role to apply my technical skills in designing reliable embedded solutions.

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PROJECTS

Smart Energy Monitoring System

- Designed a real-time energy monitoring and automation system using ESP32, ACS712 current sensors and voltage sensors with LCD and cloud dashboard integration.
- Integrated IoT automation and security features including relay control, energy analytics, EB billing estimation and remote alerts.

Smart Shopping Trolley

- Developed a Smart Shopping Trolley system using load cell sensor, and Firebase for real-time automated billing with anti-theft protection.
- Designed a mobile application with barcode scanning and digital payment integration to improve checkout efficiency and enhance payment methods in retail environments.

Multi-Wavelength Optical Microplastic Detection System

- Developed an embedded optical sensing system using multi-wavelength LEDs (UV, Visible, IR), and photodiodes to detect microplastics in water using light scattering and absorption principles.
- Implemented real-time signal processing and analysis to estimate microplastic concentration and enable portable water quality monitoring.

EDUCATION

B.E Electronics and Communication Engineering
Kongu Engineering College
Perundurai | 2024 – Present
CGPA: 8.82

Higher Secondary Education
Sri Sarada Balamandir Boys Matric. Hr. Sec. School
Salem | 2010 – 2024
Percentage: 83.83%

TECHNICAL SKILL

Programming: C, Embedded C, C++, Python, Verilog, System Verilog, VHDL

Microcontrollers: 8051, ESP32, Arduino

Sensor Interfacing: ACS712, ZMPT101B, HX711, Photodiodes, IR Sensor, LCD, Relay

Communication: UART, I2C, SPI, GPIO, PWM

PCB Skills: Proteus, Altium Designer (Basic), PCB Schematic & Layout (Basic)

Operating System: Ubuntu Linux

SOFT SKILLS

Problem Solving

Team Collaboration

Time Management

AREAS OF INTEREST

Embedded Systems

VLSI Design

CERTIFICATIONS

- PCB Design and Fabrication
- Modeling of Digital Logic Circuits Using VHDL
- Digital Design with Verilog

ACHIEVEMENTS

- Ideathon – Second Place
- Techno Fest (PoC) – Top 3
- 30 Hours Hackathon – Second Place

TOOLS

- Proteus
- Altium
- Arduino(IDE)
- Cadence (Genus, Pspice)
- ModelSim
- QuestaSim
- VS Code
- LTSpice

LANGUAGES

Kannada (Native)

Tamil (Conversational)

English (Professional)

Hindi (Beginner)